THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 02

Full Syllabus (Chapters 1–8) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** are collinear	O is the midpoint of AB
2	**(a)** $(3,\ -4)$	$\left(\frac{0+(6)}{2}, \frac{-8+(0)}{2}\right)$
3	**(a)** $(5,\ 8)$	$B = (2x_M - x_A,\ 2y_M - y_A)$
4	**(d)** lies on the line	RHS at $x = 7$ is 15, equal to 15
5	**(b)** 1 and 3	compare with $y = ax + b$
6	**(c)** 1	$x = 0.\overline{9},\ 9x = 9$
7	**(a)** -60	Brahmagupta's rules for signs
8	**(b)** 4	number of places $= \max(a, b)$ in $2^a 5^b$
9	**(c)** 40	$2ab = 40$
10	**(a)** 9801	$99^2 = (100 - 1)^2$
11	**(b)** 12 cm	half-chord $= \sqrt{10^2 - 8^2} = 6$, so chord $= 2 \times 6 = 12$
12	**(d)** 90°	angle in a semicircle
13	**(a)** 5 cm	$d = \sqrt{13^2 - 12^2} = 5$
14	**(a)** 60 cm^2	area $= \frac{1}{2}bh$
15	**(b)** 616 cm^2	$\pi r^2 = \frac{22}{7} \times 14^2 = 616$
16	**(a)** equal area	equal bases and the same height
17	**(c)** $\frac{1}{2}$	favourable outcomes: 2, 4, 6 ⇒ 3 out of 6
18	**(c)** $\frac{2}{11}$	2 favourable out of 11 letters
19	**(d)** 3^n	each black triangle becomes three
20	**(a)** 210	$\frac{20 \cdot 21}{2}$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$(0, 6)$ and $(-4, 0)$; distance $= \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}$ units.	1+1
22	$h(5) = 3 - 2.5 = 0.5$ m. Empty when $3 - 0.5t = 0 \Rightarrow t = 6$ months.	1+1
23	Divide the unit interval $[0, 1]$ into 4 equal parts and move 3 parts to the right of 0.	2
24	$(x + 5)\,(x + 6)$	2
25	$\angle A + \angle C = 180^\circ$, so by the converse (Theorem 12) the vertices are concyclic.	1+1
26	$bh = 96 \Rightarrow h = \frac{96}{12} = 8$ cm.	1+1
27	Both outcomes are equally likely ($\frac{1}{2}$ each) and no one can predict the result beforehand, so neither team has an advantage.	2
28	$a = 3, d = 5$: $t_{10} = 3 + 9 \times 5 = 48$; $t_{26} = 3 + 25 \times 5 = 128$.	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	$AB = \sqrt{9 + 16} = 5$, $AC = \sqrt{16 + 9} = 5$, $BC = \sqrt{49 + 1} = \sqrt{50}$. $AB = AC$ (isosceles) and $AB^2 + AC^2 = 50 = BC^2$ (right angle at A).	1+1+1
30	$y = 3, 5, 11, 15, 19, 25, 41$. $(4, 9)$: $2(4) + 1 = 9$ ✓ lies on the line; $(6, 14)$: $2(6) + 1 = 13 \neq 14$ ×.	2+1
31	If $\frac{2}{\sqrt{3}} = r$ (rational, $r \neq 0$) then $\sqrt{3} = \frac{2}{r}$ is rational — contradiction.	1+1+1
32	$= pqx^2 + (pb + qa)x + ab$. For $(2x + 3)(3x + 1)$: $6x^2 + (2 + 9)x + 3 = 6x^2 + 11x + 3$.	1+2
33	Radii are in the ratio $5 : 4$ (1). Areas πr^2 in the ratio $25 : 16$ (2).	1+2
34	Weekend (Sat, Sun) ⇒ $\frac{2}{7}$ (1). Days beginning with T: Tuesday, Thursday ⇒ $\frac{2}{7}$ — equally likely (2).	1+2

35	$4k + 5k = 180^\circ \Rightarrow k = 20^\circ$ (2); $\angle A = 80^\circ, \angle C = 100^\circ$ (1).	2+1
	OR: At the point where the perpendicular bisectors of the segments joining the friends meet (the circumcentre) (2). Points on a perpendicular bisector are equidistant from the ends, so this point is equidistant from all three (1).	2+1
36	Differences $2, 3, 4$ — not constant ⇒ not an AP (1+). Ratios $3, 2, \frac{5}{3}$ — not constant ⇒ not a GP (1).	1+1+1
	OR: GP with $r = 2$: $500 \times 2^6 = 32000$.	2+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) No, π is irrational. (ii) $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} = \frac{105-35+21-15}{105} = \frac{76}{105}$; times $4 = \frac{304}{105} \approx 2.895$ — a rough value, improving with more terms. (iii) Lambert, 1761.	1+2+2
	OR: (i) $\frac{126}{10} = \frac{63}{5}$. (ii) $\frac{120}{10000} = \frac{3}{250}$. (iii) $\frac{3052}{1000} = \frac{763}{250}$. (iv) $\frac{23}{99}$. (v) $x = 2.0\overline{5}$: $10x = 20.\overline{5}$, $100x = 205.\overline{5}$, $90x = 185 \Rightarrow x = \frac{37}{18}$.	1+1+1+1+1
38	(i) The arc on the other side of the diameter subtends 180° at the centre, so the inscribed angle is 90° (2). (ii) Centre: $2 \times 62^\circ = 124^\circ$ (1). ACBD is cyclic so $\angle ADB = 180^\circ - 62^\circ = 118^\circ$ (2).	2+1+2
	OR: (i) Both are angles in a semicircle: 90° each (2). (ii) $180 - 90 - 30 = 60^\circ$ (1). (iii) $180 - 90 - 50 = 40^\circ$ (1). (iv) ACBD has all four vertices on the circle, so it is cyclic (1).	2+1+1+1
39	Inscribed hexagon: six equilateral triangles of side r ⇒ perimeter $6r < 2\pi r \Rightarrow \pi > 3$ (2). Circumscribed hexagon: side $= \frac{2r}{\sqrt{3}}$ ⇒ perimeter $4\sqrt{3}r > 2\pi r \Rightarrow \pi < 2\sqrt{3} \approx 3.46$ (2). More sides squeeze the two perimeters closer, trapping π more tightly (1).	2+2+1
	OR: (i) $AB = AC = BC = r$ (radii) (1). (ii) $\angle CAB = \angle BAD = 60^\circ$, so $\angle CAD = 120^\circ$ (2). (iii) Each outer arc subtends 240° at its centre: total $= 2 \times \frac{240}{360} \times 2\pi r = \frac{8\pi r}{3}$ (2).	1+2+2
40	(i) AP with $a = 1$, last term $2n - 1$: $S = \frac{n}{2}(1 + 2n - 1) = n^2$ (2). (ii) 25 terms ⇒ $25^2 = 625$ (1). (iii) $1 + 3 + \cdots + 23 = 12^2 = 144$ (2).	2+1+2
	OR: (i) $\frac{n(n+1)}{2} + \frac{n(n-1)}{2} = \frac{n(2n)}{2} = n^2$ (2). (ii) $55 + 45 = 100$ ✓ (1). (iii) $t_{17} = 153$, $t_{80} = 3240$ (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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